Panlin Li (Direct Ph.D. Application)

High-Dimensional Statistical Learning | Data Modeling

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Member, China Society for Industrial and Applied Mathematics · Member, Shandong Young Photographers' Association · Member, Qingdao Young Photographers' Association

Education

Xinjiang Agricultural University (School of Mathematics & Physics) | Mathematics & Applied Mathematics · B.S. (Top 2.5%) | Advisor: Prof. Hua Huang | Sep. 2023 – Jul. 2027

- Honors: Outstanding Volunteer (9×) & Outstanding Team — CYL Central Committee; Hematopoietic Stem Cell Donation Honor Certificate; Outstanding Student, Innovation & Entrepreneurship Star (2 consecutive years), Science & Innovation Pioneer Scholarship — XJAU.
- Key outcomes: led/participated in 16 student innovation projects (3 national, 6 provincial), project recognized by Huawei Research Institute; first-authored 8 papers (plus 3 Q1 SCI under review); 4 utility-model patents, 4 software copyrights; 5 national & 7 provincial competition awards; volunteer work covered by *China Daily*, *People's Daily*, and other media outlets.

Research & Projects

National Statistical Science Research Project · Major Program · Under Application **Multimodal High-Dimensional Statistical Learning for Cotton Yield and Climate Risk** **2026 – Present**

- Responsibilities: lead-authored the National Major Program proposal and a 7,000-word survey under advisor guidance; proposed a heterogeneous-multimodal tensor-sparse fusion model jointly modeling meteorological / soil / remote-sensing / phenological signals; designed a spatio-temporal doubly-debiased Lasso + generalized random forest causal-identification framework; planned a multi-stream INSPECT + PCMCI field-level early warning system.
- Key outcomes: proposal submitted and under review.

Xinjiang Natural Science Foundation · General Program · Participant **Functional Statistical Modeling via Spatial-Spectral Feature Fusion of Hyperspectral Imaging** **2026 – Present**

- Responsibilities: curated multi-cultivar HSI/NIR public datasets; performed SNV/MSC correction and multi-basis functional representation; developed three algorithmic directions — functional contrastive sparse multi-task, mechanism-embedded post-harvest modeling, and residue spectral screening; lead-authored three corresponding papers.
- Key outcomes: three first-author drafts completed (under advisor review).

National Innovation Project · Ongoing · Participant **Apple Origin Traceability via Machine Learning & Transfer Learning** **2025 – Present**

- Responsibilities: lead-authored the proposal; led cross-year apple HSI and image data collection across Xinjiang/Shandong/Gansu; built a 9-domain benchmark with an origin × year double-layered leakage-controlled evaluation protocol; ran PLS-DA / SVM / 1D-CNN / Transformer baselines and cross-year generalization study; led paper writing and submission.
- Key outcomes: first-author submission to *Computers and Electronics in Agriculture* (SCI Q1 Top; IF: 10.3).

Other Projects: Small-Sample Time-Series Forecasting (National, Completed, PI) · Wenmai Zhiyuan Hub (Regional, Excellent, PI) · Shenji Miansuan — Digital Cotton Field (Regional, Ongoing, PI) · Tongxi Zhixing — Multimodal Navigation (Regional, Ongoing) · Functional Contrastive Sparse Multi-Task HSI Detection (Regional, Ongoing) · Apple SSC Prediction via FDA+Image Fusion (Regional, Excellent) · Nonlinear Wave Solutions of High-Dimensional PDEs (University, Excellent, PI) · Zhitong Smart Cane (University, Excellent) · Tianshan Pastoral Song (University, Excellent) · Non-Traveling-Wave Solutions of Nonlinear PDEs (University, Pass) · "Cure on Arrival" — AI Pest Diagnosis (University, Pass) · Online Rapid Detection via Spectral Transfer (University, Ongoing, PI) · Fruit & Vegetable Storage Quality Evolution (University, Ongoing)

Publications

- Li, Panlin; Li, Yuchang; Huang, Hua; et al. *Functional Multi-Task Sparse Learning with Spectral-Spatial Information Fusion for Hyperspectral Food Quality Detection*. Information Fusion. (SCI Q1 Top; IF: 17.4; Under Review)
- Li, Panlin; Li, Longjie; Liu, Ya; et al. *Robust cross-origin near-infrared prediction of soluble solids content in apples via a Beer–Lambert physics-informed prior*. Food Chemistry. (SCI Q1 Top; IF: 10.4; Under Review)
- Li, Panlin; Zhang, Qingli; Jiao, Qitai; et al. *Cross-year declared-origin discrimination of commercial apples by near-infrared spectroscopy: a nine-domain benchmark, a leakage-controlled evaluation protocol, and a baseline study*. Computers and Electronics in Agriculture. (SCI Q1 Top; IF: 10.3; Under Review)
- Li P, Zhang N. *Optimization of NIPT Testing Timing and Fetal Anomaly Determination Models*. International Journal of Public Health and Medical Research, 2026, 6(2): 1–8.
- Li, Panlin; Bao, Fangchen; Zhang, Jingyi; et al. *Periodic Solutions to a New Class of KdV Equations*. Advances in Applied Mathematics, 2025, 14(5): 23–28.
- Li P, Zhang N. *Prediction of optimal NIPT timing and analysis of influencing factors of fetal Y chromosome concentration based on robust Bayesian ridge regression*. Academic Journal of Computing & Information Science, 2025, 8(10): 99–107.
- Li P, Zhang N. *Study on the determination of fetal chromosomal abnormalities based on multi-model regression*. Academic Journal of Mathematical Sciences, 2025, 6(2): 121–129.
- Li P, Zhang N. *Study on the effect of n-hexane insolubles on pyrolysis product yields based on statistical regression analysis*. Highlights in Science, Engineering and Technology, 2024, 116: 270–276.
- Li P, Zhang N. *Research on Optimizing High-dimensional Data Model Based on Principal Component Analysis*. In: 2025 IEEE 3rd International Conference on Control, Electronics and Computer Technology (ICCECT), Jilin, China, 2025, pp. 998–1004.

- Li, Panlin; Li, Jie; Alamusijiang Kadir; et al. *Cultural Heritage and Rural Revitalization: Exploration and Reflections on China's Practice*. Smart Agriculture Guide, 2024, 4(21): 185–188.
- Rao W J, Li P L. *An improved kernel regularized nonhomogeneous grey model based on conformable fractional accumulation*. Transactions on Computational and Applied Mathematics, 2024, 4: 137–147. (Co-First Author)
- Zhang, Ning; Li, Panlin; Bao, Fangchen; Liu, Xiaoyu. *Solitary Wave Solutions and Interaction Solutions of an Extended (3+1)-Dimensional Shallow Water Wave Equation*. Journal of Changchun Normal University, 2026, 45(4): 6–11.
- Jiao, Qitai; Li, Panlin; Mei, Tao; et al. *ESP32-Based Multi-Sensor Near- & Far-Range Water Quality Monitoring System*. Internet of Things Technologies, 2026, 16(2): 9–12, 16.
- Huang, Hua; Wang, Mo; Liu, Ya; Li, Panlin; et al. *Prediction of Apple Soluble Solids Content Using a Functional Partial Linear Regression Model Fusing Image Color Features and Vis-NIR Spectra*. Spectroscopy and Spectral Analysis, 2025, 45(S1): 415–423.
- **Patents (4, granted):** Section-Connection Component for a White Cane (CN2024230029031) · Portable Ultrasonic Detection Device (CN2024230026103) · Pesticide Spraying Drone (CN20241226759) · Foldable Anti-Collision Spraying Boom for Drones (CN20241226785).
- **Software Copyrights (4):** Intelligent White Cane Vision Recognition System V1.0 · Smart Cane Data Collection & Monitoring System V1.0 · User Behavior Analysis and Recommendation System V1.0 · Big-Data-Driven Market Analysis and Forecasting System V1.0.

Competitions & Awards

- "Chia Tai Cup" 16th National College Students Market Survey & Analysis Contest · Undergraduate Third Prize *Team Lead* 2026.05
- 11th National Undergraduate Statistical Modeling Contest · Undergraduate Third Prize *Team Lead* 2025.10
- "Chia Tai Cup" 15th National College Students Market Survey & Analysis Contest · Undergraduate Third Prize *Team Lead* 2025.05
- China International College Students' Innovation Competition (2024) · Bronze Award *Participant* 2024.10
- DataBee Cup National College Student Interview Survey Contest · Third Prize *Participant* 2024.10
- 14th China Innovation & Entrepreneurship Competition Xinjiang Division (12th Xinjiang Competition) · Growth Group Third Prize *Team Lead* 2025.10
- China International College Students' Innovation Competition (2025) Xinjiang Division · Bronze Award *Team Lead* 2025.08
- 11th National Undergraduate Statistical Modeling Contest Xinjiang Division · Undergraduate First Prize *Team Lead* 2025.08
- "Chia Tai Cup" 15th National College Students Market Survey & Analysis Contest Xinjiang Division · Undergraduate First Prize *Team Lead* 2025.04
- 2024 Xinjiang Division Information Literacy Competition · Excellence Award *Team Lead* 2024.12
- 2024 National College Students Mathematical Modeling Contest Xinjiang Division · Undergraduate Second Prize *Team Lead* 2024.11
- 10th National Undergraduate Statistical Modeling Contest Xinjiang Division · Second Prize *Team Lead* 2024.07

Student Experience

Class Monitor, Mathematics 2302, School of Mathematics & Physics 2023.09 – 2027.07
Class Monitor, 19th Youth Marxist Training Program, Xinjiang Agricultural University 2025.05 – 2026.05
Head, Volunteer Association, School of Mathematics & Physics, XJAU 2024.09 – 2025.11

- Organized academic and sports activities; led the class to receive *Outstanding Class Collective* and *Sports Pioneer Class* honors (AY 2024–2025).
- Founded the "Shield of the Future" volunteer team (20 members); participated in 11 CYL Central Committee / CYVA public-welfare programs across Shandong, Henan, and Xinjiang; work reported by *China Daily*, *People's Daily*, and other media outlets.

Skills

- **Language:** TOEFL 110/120; first-authored multiple English SCI/IEEE papers.
- **Professional:** high-dimensional statistical inference (debiased Lasso, double ML, generalized random forests), functional data analysis (FPCA, function-on-function regression), tensor decomposition & multimodal fusion; Python (PyTorch / scikit-learn), end-to-end modeling.
- **Tools:** LaTeX, Git, Lightroom / Final Cut Pro.

Self-Assessment

- Photography, freedom, the great outdoors; badminton (enthusiastic but mediocre); can hold a drink (never been drunk). Love diving into new things, eager to figure stuff out — strong hands-on, execution, and information-synthesis skills, driven by a deep exploratory urge.
- Through 16 projects and 11 papers during undergrad, I have walked the full loop from math → statistics → ML → real-world deployment — which has made me clearly aware that my bottleneck lies in theoretical depth. In the Ph.D. stage, I aim to move toward more theoretical directions while staying grounded in real-world applications, finding balance between methodological rigor and practical impact.